



UNIVERSITY OF
GEORGIA

College of Agricultural &
Environmental Sciences

SNP effects from ssGBLUP using the BLUPF90 family (postGSf90)

Equivalence between GBLUP and SNP-BLUP

GBLUP

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{W} \\ \mathbf{W}'\mathbf{X} & \mathbf{W}'\mathbf{W} + \mathbf{G}^{-1}\lambda_1 \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{W}'\mathbf{y} \end{bmatrix}$$

$$\text{Var}(\mathbf{u}) = \mathbf{G}\sigma_u^2$$

SNP-BLUP

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \mathbf{I}\lambda_2 \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{a}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix}$$



SNP effects

$$\mathbf{u} = \mathbf{Z}\mathbf{a}$$

Are GBLUP and SNP-BLUP equivalent?

- Assumption of GBLUP: $\text{Var}(\mathbf{u}) = \mathbf{G}\sigma_u^2$
- In SNP-BLUP: $\mathbf{u} = \mathbf{Z}\mathbf{a}$

$$\mathbf{u} = \mathbf{Z}\mathbf{a}$$

$$\text{Var}(\mathbf{u}) = \text{Var}(\mathbf{Z}\mathbf{a})$$

$$\text{Var}(\mathbf{u}) = \mathbf{Z} \text{Var}(\mathbf{a}) \mathbf{Z}'$$

$$\text{Var}(\mathbf{u}) = \mathbf{Z}\mathbf{Z}'\sigma_a^2$$

$$\sigma_a^2 = \frac{\sigma_u^2}{2 \sum_{i=1}^{SNP} p_i(1 - p_i)}$$

$$\text{Var}(\mathbf{u}) = \mathbf{Z}\mathbf{Z}' \frac{\sigma_u^2}{2 \sum_{i=1}^{SNP} p_i(1 - p_i)}$$

$$\text{Var}(\mathbf{u}) = \boxed{\frac{\mathbf{Z}\mathbf{Z}'}{2 \sum_{i=1}^{SNP} p_i(1 - p_i)}} \sigma_u^2$$

Genomic
relationship matrix
VanRaden (2008)

$$\mathbf{G} = \frac{\mathbf{Z}\mathbf{Z}'}{2 \sum_{i=1}^{SNP} p_i(1 - p_i)}$$

$$\text{Var}(\mathbf{u}) = \mathbf{G}\sigma_u^2$$



GBLUP assumption!!!

GBLUP and SNP-BLUP are equivalent!

If we can get $\mathbf{u}$ ($\mathbf{u} = \mathbf{Za}$) from SNP-BLUP, we can get $\mathbf{a}$ from GBLUP!



Review

Single-Step Genomic Evaluations from Theory to Practice: Using SNP Chips and Sequence Data in BLUPF90

Daniela Lourenco ^{1,*}, Andres Legarra ², Shogo Tsuruta ¹ , Yutaka Masuda ¹, Ignacio Aguilar ³ 
and Ignacy Misztal ¹

<https://www.mdpi.com/2073-4425/11/7/790>

ssGBLUP and ssSNP-BLUP are also equivalent!

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{W} \\ \mathbf{W}'\mathbf{X} & \mathbf{W}'\mathbf{W} + \mathbf{H}^{-1} \frac{\sigma_e^2}{\sigma_u^2} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{W}'\mathbf{y} \end{bmatrix}$$

ssGBLUP

Misztal et al. (2009)
Legarra et al. (2009)
Aguilar et al. (2010)
Christensen & Lund (2010)

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z}\mathbf{M} & \mathbf{X}'_n\mathbf{Z}_n \\ \mathbf{M}'\mathbf{Z}'\mathbf{X} & \mathbf{M}'\mathbf{Z}'\mathbf{Z}\mathbf{M} + \mathbf{I} \frac{\sigma_e^2}{\sigma_a^2} & \mathbf{M}'_n\mathbf{Z}'_n\mathbf{Z}_n \\ \mathbf{Z}'_n\mathbf{X}_n & \mathbf{Z}'_n\mathbf{Z}_n\mathbf{M}_n & \mathbf{Z}'_n\mathbf{Z}_n + \mathbf{A}^{nn} \frac{\sigma_e^2}{\sigma_g^2} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\alpha}} \\ \hat{\boldsymbol{\epsilon}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{M}'\mathbf{Z}'\mathbf{y} \\ \mathbf{Z}'_n\mathbf{y}_n \end{bmatrix}$$

ssSNPBLUP or ssBR

Fernando et al. (2014)
Liu et al. (2014)
Mantysaari & Strandén (2016)



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<https://doi.org/10.3168/jds.2018-14913>

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Short communication: Genomic prediction using different single-step methods in the Finnish red dairy cattle population

H. Gao,^{*,†} M. Koivula,[‡] J. Jensen,^{*} I. Strandén,[‡] P. Madsen,^{*} T. Pitkänen,[‡] G. P. Aamand,[‡] and E. A. Mäntysaari[‡]

^{*}Center for Quantitative Genetics and Genomics, Department of Molecular Biology and Genetics, Aarhus University, DK-8830 Tjele, Denmark

[†]Nordic Cattle Genetic Evaluation, DK-8200 Aarhus, Denmark

[‡]Natural Resources Institute Finland (Luke), FIN-31600 Jokioinen, Finland

We confirmed that regular ssGBLUP and ssBR with an extra polygenic effect led to the same predictions.

SNP effects in ssGBLUP

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{W} \\ \mathbf{W}'\mathbf{X} & \mathbf{W}'\mathbf{W} + \mathbf{H}^{-1}\lambda_1 \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{W}'\mathbf{y} \end{bmatrix}$$

$$\hat{\mathbf{a}} = \alpha b \frac{1}{2\sum p_i(1-p_i)} \mathbf{Z}' \mathbf{G}^{-1} \hat{\mathbf{u}}$$

Matrix of SNP content
Genomic relationship matrix

α = blending parameter for $\mathbf{G}$

$$\lambda = \frac{1}{n^2} \left(\sum_i \sum_j \mathbf{A}_{22ij} - \sum_i \sum_j \mathbf{G}_{ij} \right) \quad b = 1 - \frac{\lambda}{2}$$

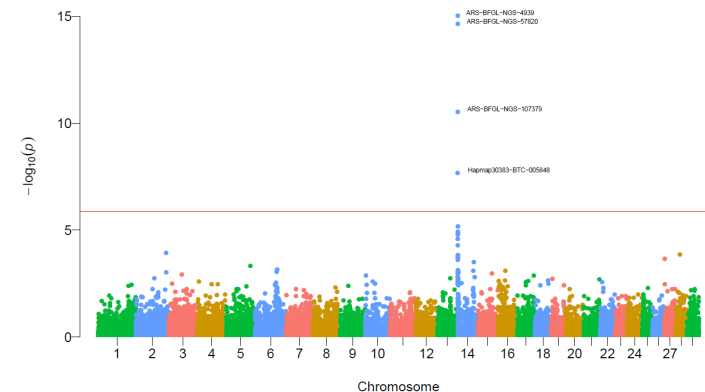
What can we do with SNP effects?

1) Predictions for animals not included in the evaluation

Indirect predictions

Indirect Genomic Predictions

2) Genome-Wide Association Studies (GWAS)



1) Indirect predictions

- Interim evaluations
 - Between official runs
- Not all genotyped animals are in the evaluations
 - Animals with incomplete pedigree increase bias and lower R^2
- Commercial products
 - e.g. GeneMax for non-registered animals

1) Indirect predictions

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{W} \\ \mathbf{W}'\mathbf{X} & \mathbf{W}'\mathbf{W} + \mathbf{H}^{-1}\lambda_1 \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{W}'\mathbf{y} \end{bmatrix} \quad \Longrightarrow \quad \hat{\mathbf{a}} = \alpha b \frac{1}{2\sum p_i(1-p_i)} \mathbf{Z}'\mathbf{G}^{-1}\hat{\mathbf{u}}$$

Indirect Prediction: $\mathbf{IP} = \mathbf{u}_m^* = \mathbf{Z}\hat{\mathbf{a}}$

1) Indirect predictions

Indirect Prediction: $\mathbf{u}_m^* = \mathbf{Z}\hat{\mathbf{a}}$

└─ Fine if comparing among animals with IP

- Not fine if comparing it with GEBV from the main evaluation
 - Put in the pedigree scale

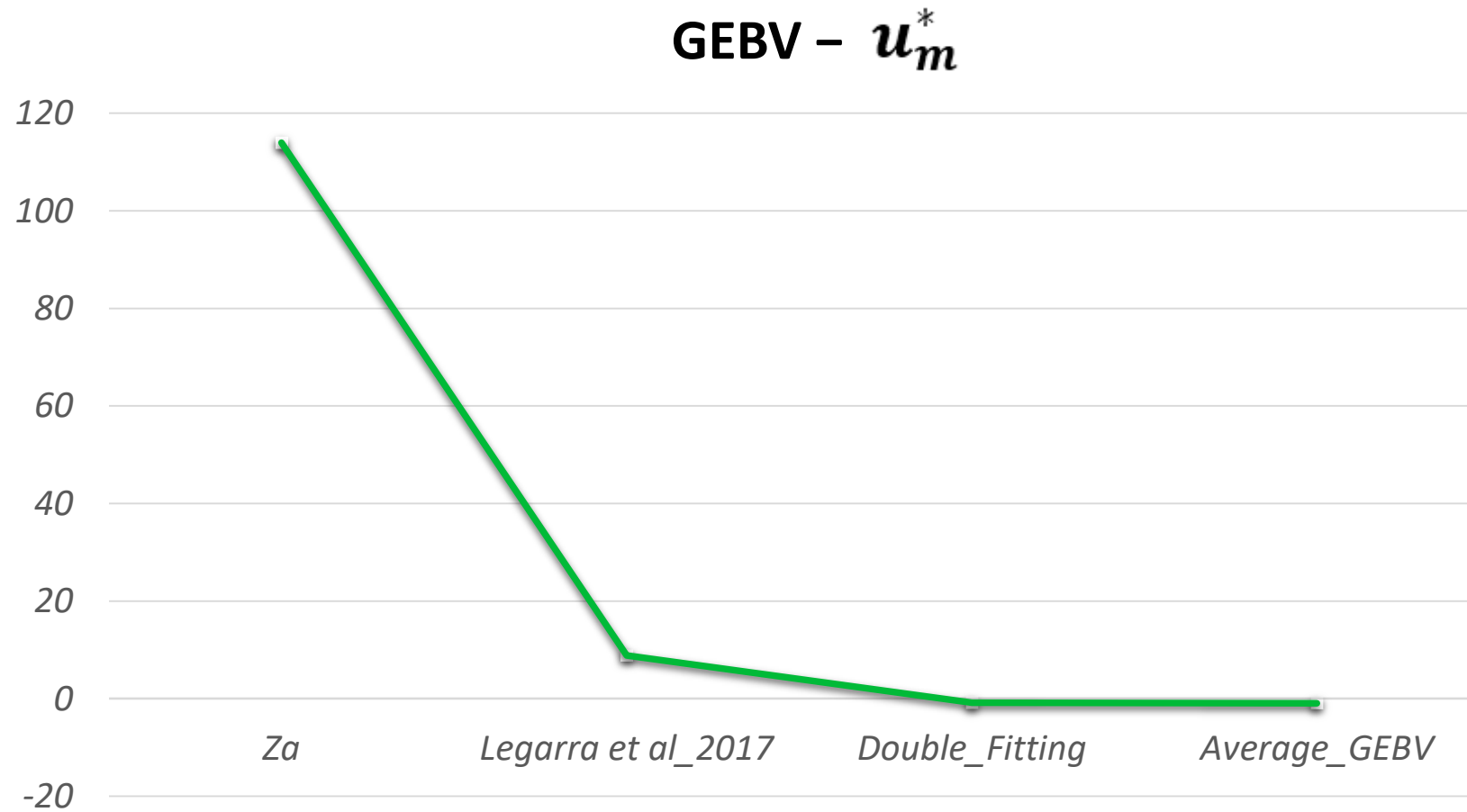
$$\mathbf{u}_m = \hat{\boldsymbol{\mu}} + \mathbf{u}_m^*$$

└─ $\hat{\boldsymbol{\mu}} = \alpha\lambda\mathbf{1}'\mathbf{G}^{-1}\hat{\mathbf{u}}$

α = blending parameter for $\mathbf{G}$

$$\lambda = \frac{1}{\mathbf{n}^2} \left(\sum_i \sum_j \mathbf{A}_{22ij} - \sum_i \sum_j \mathbf{G}_{ij} \right)$$

1) Indirect predictions



How to compute Indirect predictions

1) Pedigree + phenotypes + genotypes

2) **renumf90**

3) preGSf90 to save clean files

4) **blupf90+** (save the clean files)

- Good practice to save time: `OPTION saveGInverse + OPTION saveA22Inverse`

5) **postGSf90** (with clean files)

- BLUPF90 family software to compute SNP effects (+more)
- Same parameter file as `blupf90+`
- Good practice to save time: `OPTION readGInverse + OPTION readA22Inverse`

Output from postGSf90

snp_sol

<http://nce.ads.uga.edu/wiki/doku.php?id=readme.pregsf90>

contains solutions of SNP and weights

- 1: trait
- 2: effect
- 3: SNP
- 4: Chromosome
- 5: Position
- 6: SNP solution
- 7: weight

snp_pred

- 1st line: model, tuning, blending information
- 2nd line: Trait/effect infor
- AF in 10 columns
- $\mu_{\hat{}}$, $\text{var}_{\hat{\mu}}$
- SNP effects

How to compute Indirect predictions

6) `predf90`

- Have to provide a SNP file for the new genotyped animals to receive IP
 - same SNP as in the clean file

```
predf90 --snpfile newgen.txt --use_mu_hat
```

- The last statement adds the base, so that we have: $\mathbf{u}_m = \hat{\boldsymbol{\mu}} + \mathbf{u}_m^*$

Output from pred90

SNP_predictions

Animal ID	SNP call rate	Indirect Predictions
-----------	---------------	----------------------

UGA50014	1.00	0.17414457
UGA50016	1.00	0.72332874E-01
UGA50042	1.00	1.0016705
UGA50058	1.00	0.17190497
UGA50060	1.00	0.98674759E-01
UGA50065	1.00	-0.60623702E-01
UGA50073	1.00	-0.17860851
UGA50077	1.00	-0.21597147
UGA50079	1.00	-0.69586390
UGA50084	1.00	1.0600574
UGA50085	1.00	-0.28602412
UGA50088	1.00	-0.12758011

pred90 can also compute accuracy of indirect predictions

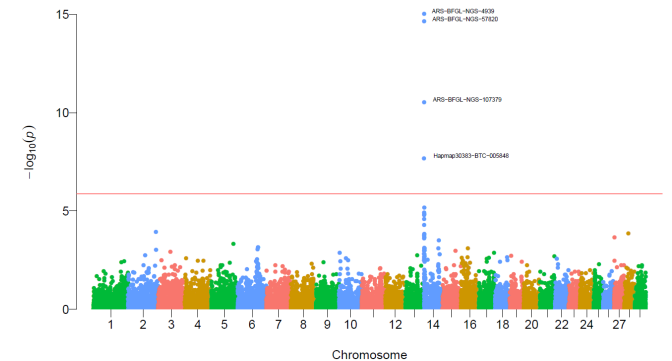
OPTION snp_p_value #in blupf90+
OPTION snp_var #in postGSf90
--acc #in pred90

<http://nce.ads.uga.edu/wiki/doku.php?id=readme.pred90>

2) Genome-wide Association Studies

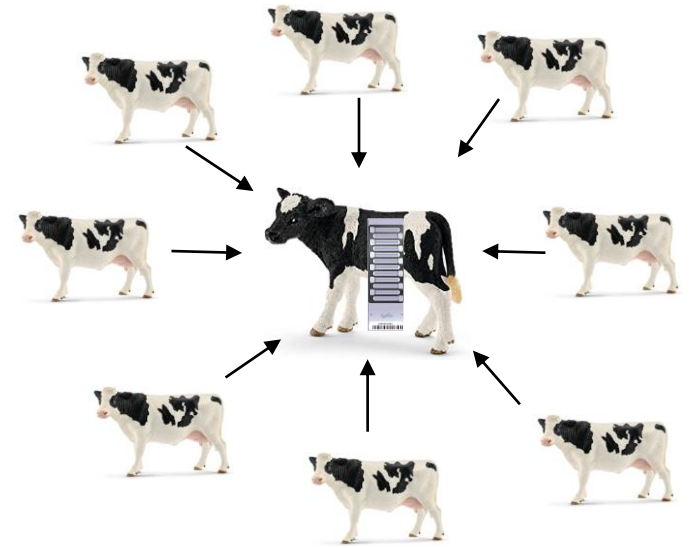
Current standard for GWAS

- Single marker regression with **G** to compensate for relationships
 - $\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{z}_i\mathbf{a}_i + \mathbf{u} + \mathbf{e}$
 - **z**: gene content {0,1,2}
 - **a**: SNP effect
- Estimate SNP effects
- Get p-values as $pval_i = 2 \left(1 - \Phi \left(\left| \frac{\hat{a}_i}{sd(\hat{a}_i)} \right| \right) \right)$
- Apply Bonferroni to correct for multiple testing
- **Assumption: Genotyped individuals have phenotypes**



GWAS in livestock populations

- Most animals are non-genotyped
- Animals may not have phenotypes
- Some traits are sex-limited
 - milk, fat, protein
- Single marker regression
 - Only genotyped animals with phenotypes
 - Deregressed EBV
- Need a method that fits the livestock data
 - ssGWAS



Single-step GWAS (historical)

SNP
effects

GEBVs

$$\hat{\mathbf{a}} = \alpha b \frac{1}{2\sum p_i(1-p_i)} \mathbf{Z}' \mathbf{G}^{-1} \hat{\mathbf{u}}$$

VanRaden 2008
Stranden and Garrick 2009
Wang et al. 2012

a) Quadratic SNP variance (Falconer & Mackay, 1996)

$$d_i = \hat{a}_i^2 2p_i(1-p_i)$$

b) Nonlinear SNP variance (VanRaden, 2008)

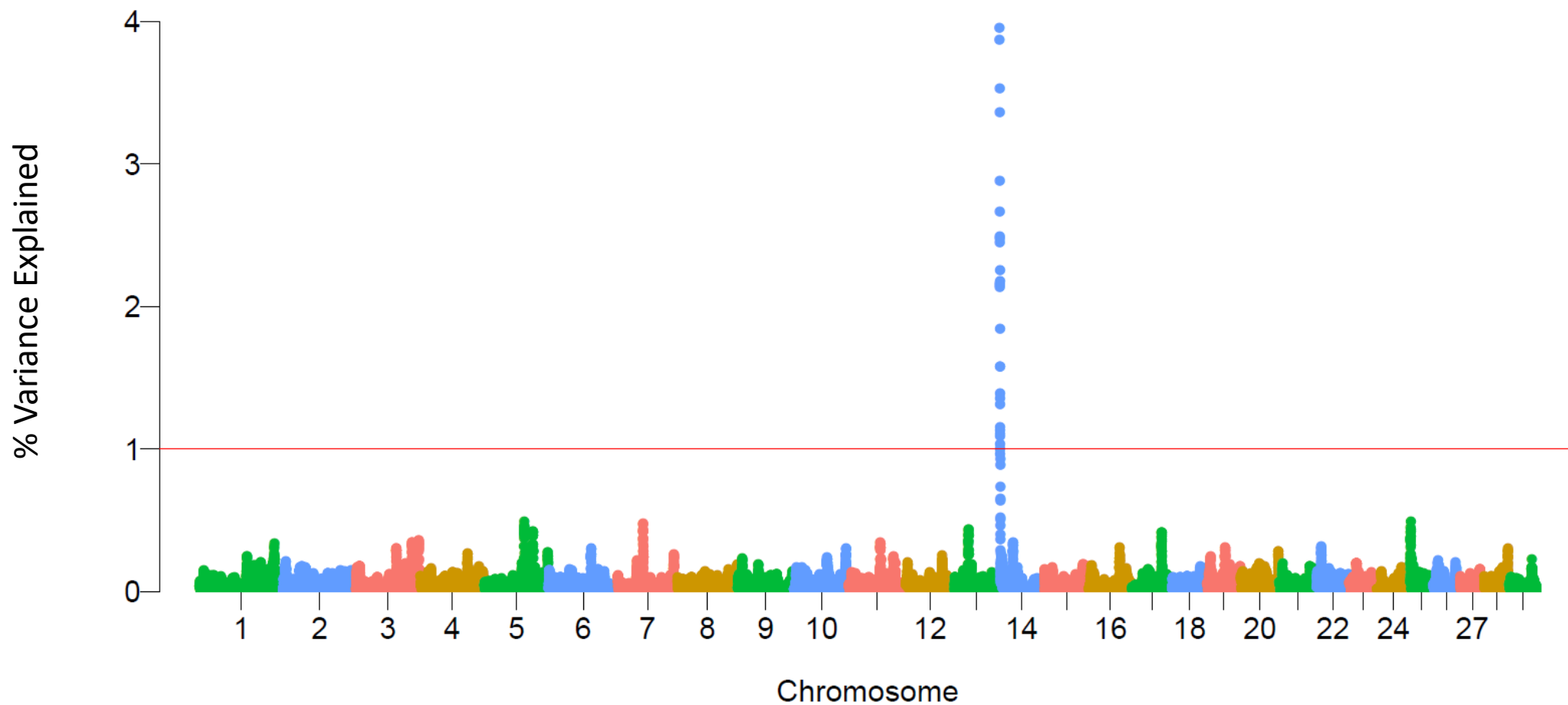
$$d_i = 1.125 \frac{|\hat{a}_i|}{sd(\hat{\mathbf{a}})} - 2$$

Single-step GWAS

Fat – US Holsteins

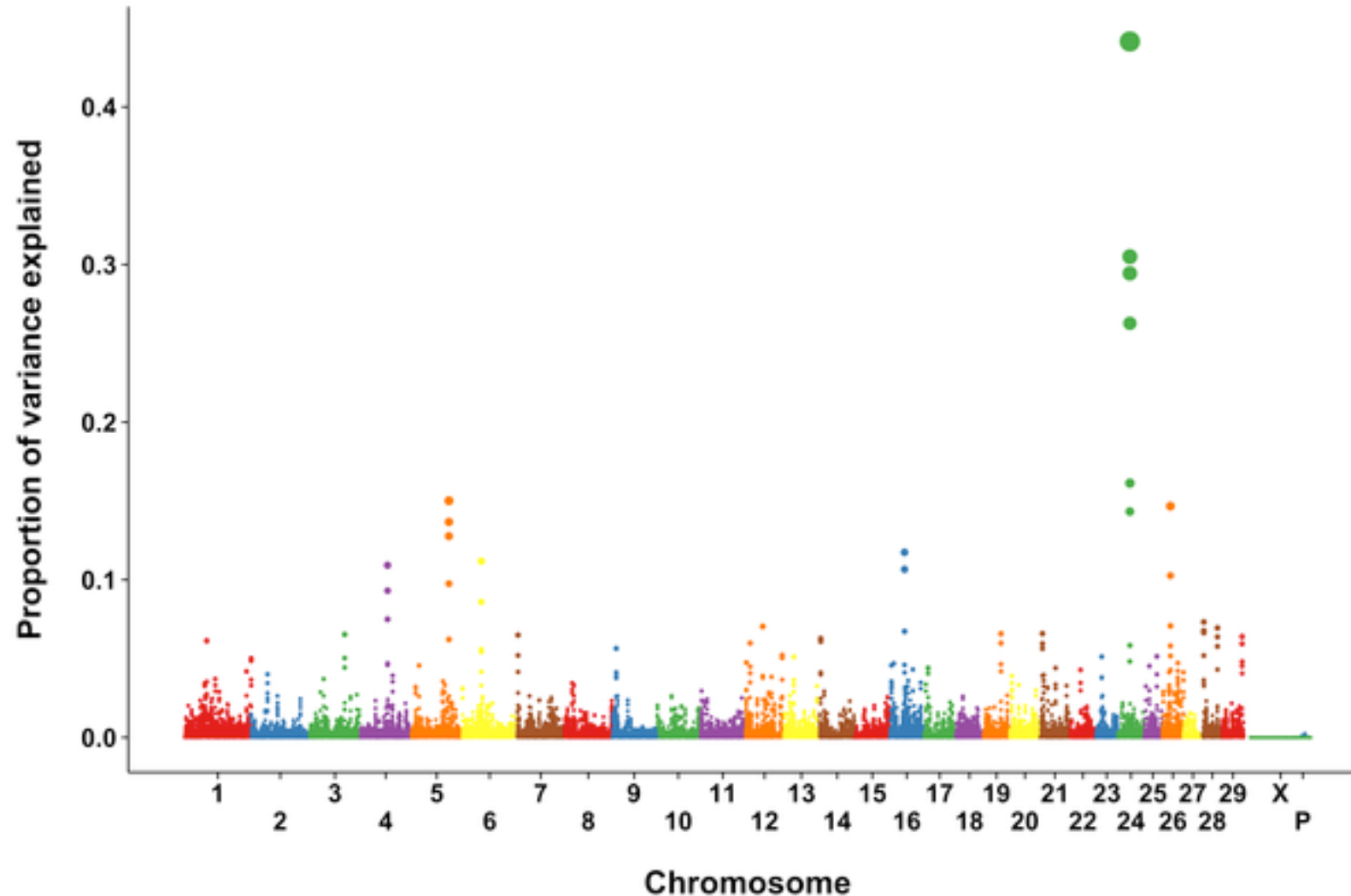
No P-value!!!

Manhattan plot of Variances



Single-step GWAS

Figure 2. Proportion of SNP variance explained by 5-SNP moving windows for rectal temperature from a **single-step GBLUP** analysis



No P-value!!!

Can we have p-values in ssGWAS?

Gualdrón Duarte et al. *BMC Bioinformatics* 2014, 15:246
<http://www.biomedcentral.com/1471-2105/15/246>



METHODOLOGY ARTICLE

Open Access

Rapid screening for phenotype-genotype associations by linear transformations of genomic evaluations

Jose L Gualdrón Duarte¹, Rodolfo JC Cantet¹, Ronald O Bates², Catherine W Ernst², Nancy E Raney² and Juan P Steibel^{2,3*}

Genome-Wide Association Analyses Based on Broadly Different Specifications for Prior Distributions, Genomic Windows, and Estimation Methods

Chunyu Chen,¹ Juan P. Steibel, and Robert J. Tempelman
Department of Animal Science, Michigan State University, East Lansing, Michigan 48824
ORCID ID: 0000-0002-7833-6730 (R.J.T.)

Aguilar et al. *Genet Sel Evol* (2019) 51:28
<https://doi.org/10.1186/s12711-019-0469-3>

SHORT COMMUNICATION



Open Access

Frequentist p-values for large-scale-single step genome-wide association, with an application to birth weight in American Angus cattle

Ignacio Aguilar¹, Andres Legarra^{2*}, Fernando Cardoso^{3,4}, Yutaka Masuda⁵, Daniela Lourenco⁵ and Ignacy Misztal⁵



doi: 10.1111/age.12378

Meta-analysis of genome-wide association from genomic prediction models

Y. L. Bernal Rubio^{*†}, J. L. Gualdrón Duarte^{*}, R. O. Bates^{*}, C. W. Ernst^{*}, D. Nonneman[‡], G. A. Rohrer[‡], A. King[‡], S. D. Shackelford[‡], T. L. Wheeler[‡], R. J. C. Cantet^{†§} and J. P. Steibel^{*¶}



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<https://doi.org/10.3168/jds.2017-13364>
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Genome-wide association analyses based on a multiple-trait approach for modeling feed efficiency

Y. Lu,^{*} M. J. Vandehaar,^{*} D. M. Spurlock,[†] K. A. Weigel,[‡] L. E. Armentano,[‡] E. E. Connor,[§] M. Coffey,[#] R. F. Veerkamp,^{||} Y. de Haas,^{||} C. R. Staples,[¶] Z. Wang,^{**} M. D. Hanigan,^{††} and R. J. Tempelman^{*†}

P-values in ssGWAS

1) Factorize and Invert LHS of ssGBLUP with YAMS (Masuda et al., 2014)

2) Solve the MME for $\begin{bmatrix} \hat{\beta} \\ \hat{u} \end{bmatrix}$ using the sparse Cholesky factor

3) Extract coefficients for genotyped animals ($\mathbf{C}^{u_2 u_2}$) from LHS⁻¹

4) Obtain individual prediction error variance of SNP effects:

$$Var(\hat{a}_i) = \alpha b \frac{1}{2 \sum p_i (1 - p_i)} \mathbf{z}_i' \mathbf{G}^{-1} (\mathbf{G} \sigma_u^2 - \mathbf{C}^{u_2 u_2}) \mathbf{G}^{-1} \mathbf{z}_i \frac{1}{2 \sum p_i (1 - p_i)} \alpha b$$

(Gualdron-Duarte et al., 2014)

5) Backsolve GEBV to SNP effects ($\hat{a}$): $\hat{a} = \alpha b \frac{1}{2 \sum p_i q_i} \mathbf{Z}' \mathbf{G}^{-1} \hat{u}$

$$6) p\text{-value}_i = 2 \left(1 - \Phi \left(\left| \frac{\hat{a}_i}{sd(\hat{a}_i)} \right| \right) \right)$$

Φ is the cumulative standard normal function

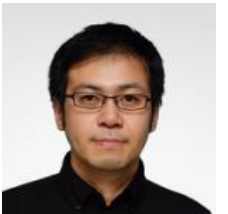
blupf90+



Ignacio
Aguilar



Andres
Legarra



Yutaka
Masuda

postGSf90

How to run ssGWAS with p-values in BLUPF90

- After renumf90 and preGSf90 to save clean files:
 - blupf90+ to estimate GEBV
 - OPTION SNP_file `snp.dat_clean`
 - OPTION map_file `mrkmap.txt_clean`
 - OPTION saveGINverse
 - OPTION saveA22Inverse
 - OPTION snp_p_value
 - postGSf90 to backsolve GEBV to SNP effect
 - OPTION SNP_file `snp.dat_clean`
 - OPTION map_file `mrkmap.txt_clean`
 - OPTION readGINverse
 - OPTION readA22Inverse
 - OPTION snp_p_value

Output from postGSf90

chr_{sn}_pval

contains data to create plot by GNUPLOT

- 1: trait
- 2: effect
- 3: $-\log_{10}(\text{p-value})$
- 4: SNP
- 5: Chromosome
- 6: Position in bp

Pft1e2.gnuplot

Pft1e2.R

chr_{sn}

contains data to create plot by GNUPLOT

- 1: trait
- 2: effect
- 3: values of SNP effects to use in Manhattan plots $\rightarrow [\text{abs}(\text{SNP}_i)/\text{var}(\text{SNP})]$
- 4: SNP
- 5: Chromosome
- 6: Position

Sft1e2.gnuplot

Sft1e2.R

Output from postGSf90

snp_sol

contains solutions of SNP and weights

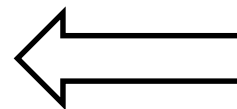
- 1: trait
- 2: effect
- 3: SNP
- 4: Chromosome
- 5: Position
- 6: SNP solution
- 7: weight

if `OPTION windows_variance` is used

- 8: variance explained by n adjacents SNP.

if `OPTION snp_p_value` is used

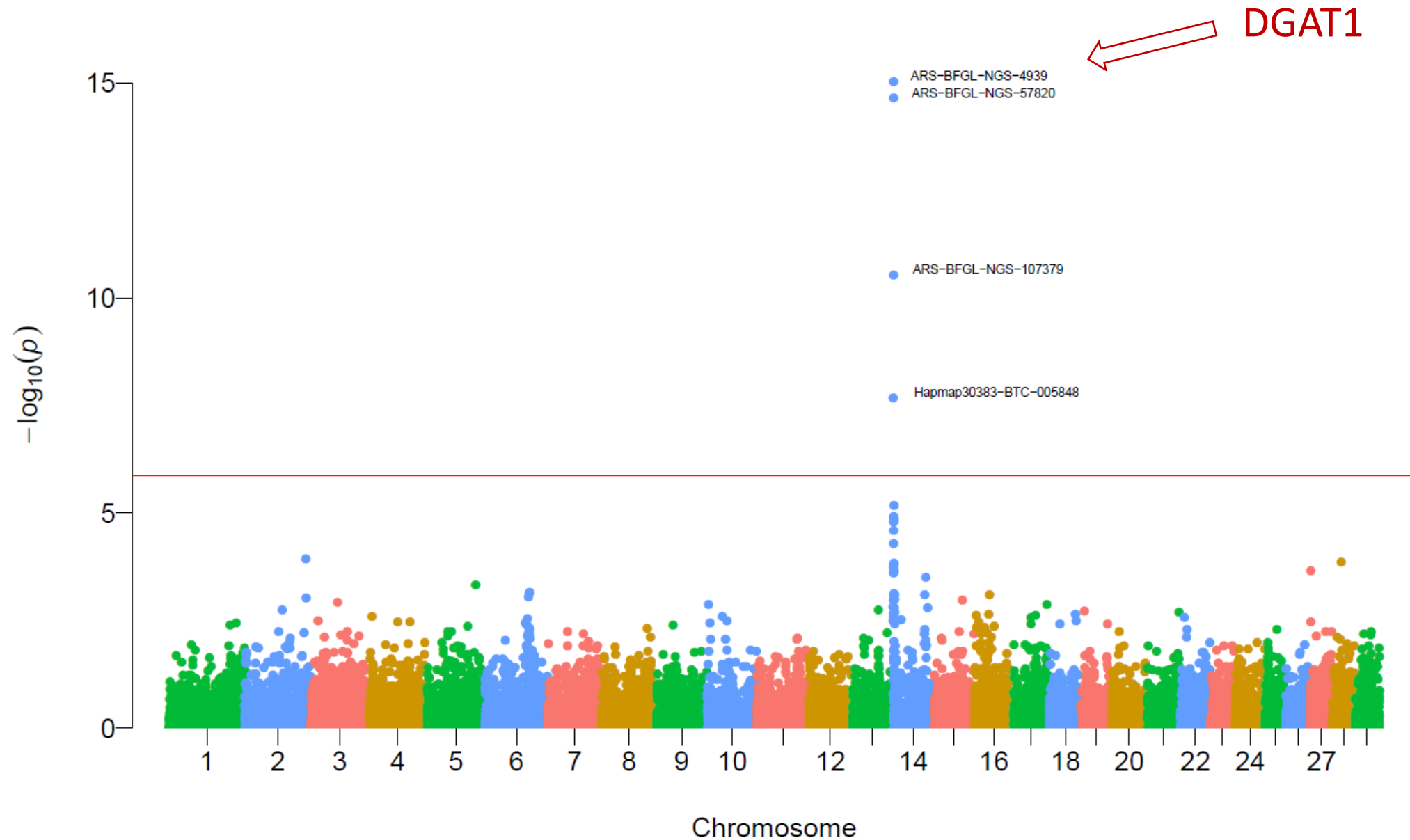
- 9: variance of the SNP solution (used to compute the p-value)



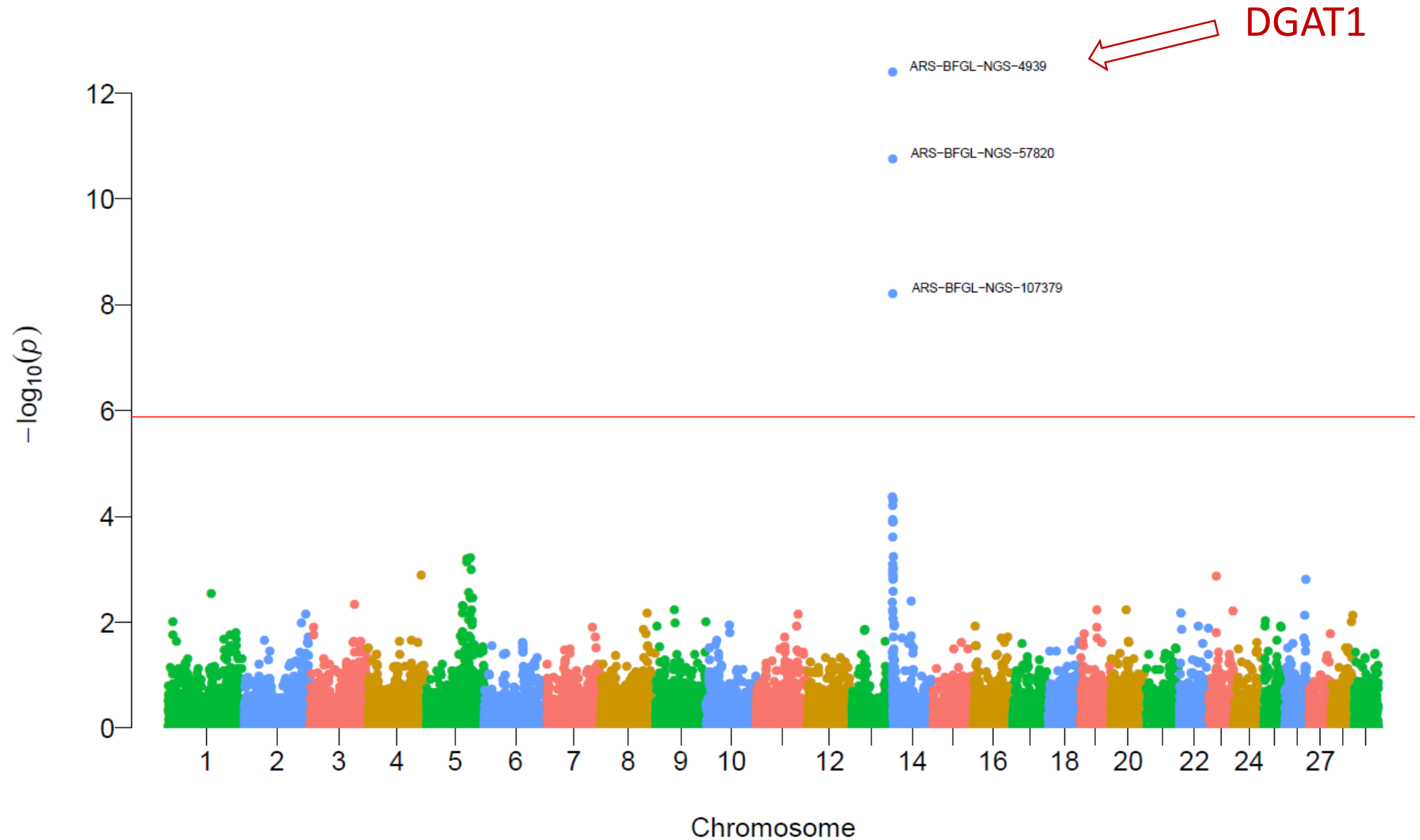
P-values in ssGWAS for US Holsteins

- US HOL 2009 data: milk, fat, protein
- Single-trait models
 - 10k genotyped bulls
 - 752k records for 100k daughters
 - 303k animals in ped

P-values in ssGWAS - Milk

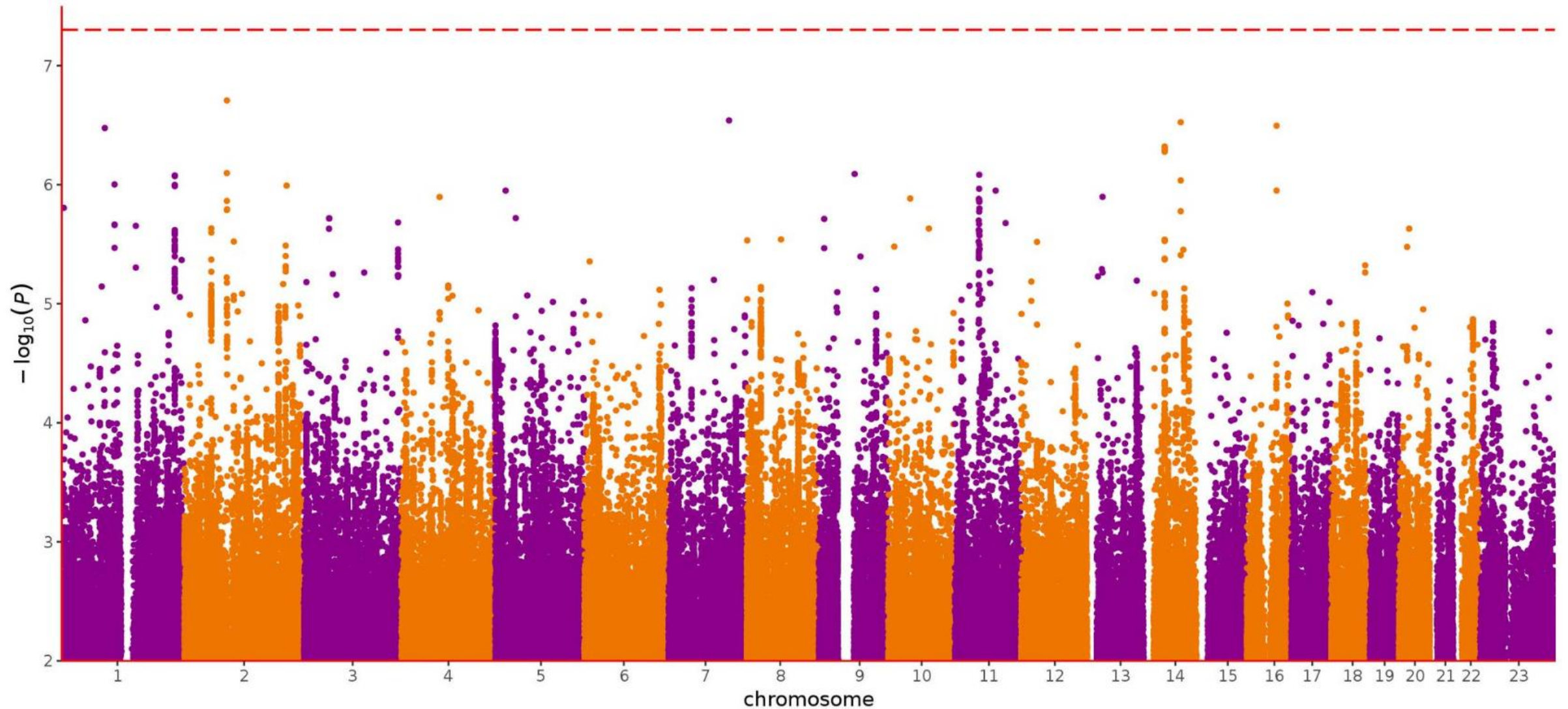


P-values in ssGWAS - Fat



Non-significant hits

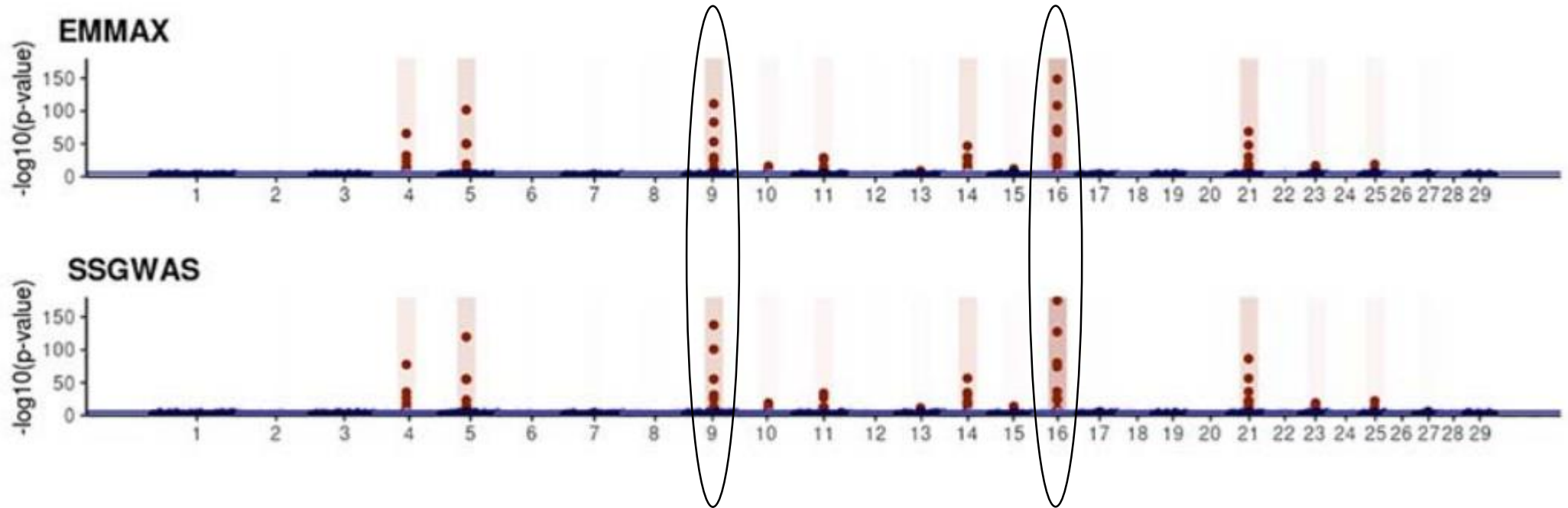
Work/job satisfaction
N=82190



ssGWAS vs. EMMAX

- Simulated population (1 QTN per CHR)

14k genotyped sires
Deregressed EBV
(10 daughters)



14k genotyped sires
500k Pedigree
250k phenotypes

Association	EMMAX (Khang et al., 2010)	ssGWAS (Aguilar et al., 2019)
True Positive	55.2 ^a (3.7)	61.6 ^a (8.7)
False Positive	0.0	0.0

postGSf90 options

<http://nce.ads.uga.edu/wiki/doku.php?id=readme.pregsf90>

OPTION Manhattan_plot

Uses GNUPLOT to plot the Manhattan plot (SNP effects) for each trait and correlated effect.

OPTION Manhattan_plot_R

Uses R to plot the Manhattan plot (SNP effects) for each trait and correlated effect.

pdf images are created: *manplot_St1e2.pdf*, but other formats can be specified.

Note: *t1e2* corresponds to trait 1, effect 2.

OPTION Manhattan_plot_R_format <format>

Control the format type to create images in R

`format` values accepted:

- pdf (default)
- png
- tif

OPTION plotsnp <n>

Control the values of SNP effects to use in Manhattan plots

- 1: plot regular SNP effects: `abs(val)`
- 2: plot standardized SNP effects: `abs(val/sd)` (default)

BLUPF90 Family of Programs

Now with support for genomic selection

Ignacy Misztal and collaborators, University of Georgia

BLUPF90 family of programs is a collection of software in Fortran 90/95 for mixed model computations in animal breeding. The goal of the software is to be as simple as with a matrix package and as efficient as in a programming language. For general description, see a [paper](#) from the CCB'99 workshop or see a [paper](#) on BGF90 at 7th WCGALP.

For variance component estimation, the family offers choices for simple and complicated models; see paper ["Reliable computing in estimation of variance components"](#). From 2009 the programs are successively modified for genomic selection using a [single-step](#) approach (or ssGBLUP) by Ignacio Aguilar and Shogo Tsuruta.

For support, join [blupf90](#) group at yahoo.com.

Troubleshooting

⚠ If the software crashes with segmentation fault, please change settings in your operating system. See [FAQ:Segmentation fault](#) for details. Also, The [FAQ pages](#) provide useful suggestions and solutions.

Headline

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